

6.3

12. a) $\binom{12}{3}$ b) $\binom{12}{0} + \binom{12}{1} + \binom{12}{2} + \binom{12}{3}$

c) $2^{12} - \binom{12}{0} - \binom{12}{1} - \binom{12}{2}$ d) $\binom{12}{6}$

22. a) $7!$ (treat ED as a unit)

c) $5!$ (treat BA and FGH as units)

e) $4!$ (treat ABCDE as a string)

24. There are $\binom{11}{6}$ ways to choose positions for men amongst the women, then $10!$ ways to order women and $6!$ ways to order men.
Thus $\binom{11}{6} 10! 6!$ in total.

36. $\binom{10}{4} \binom{15}{2} + \binom{10}{5} \binom{15}{1} + \binom{10}{6} \binom{15}{0}$

Graphs: Each of the $\binom{n}{2}$ pairs of vertices may or may not have an edge connecting them. Hence there are $2^{\binom{n}{2}}$ graphs on n distinct vertices.

6.4

5. 101 terms

12a. $x^6 y^9$ occurs with 3 $5x^2$ terms and 3 $2y^3$ terms, so the coefficient is $\binom{6}{3} \cdot 5^3 \cdot 2^3$.

14. By the binomial theorem, $(x + \frac{1}{x})^{100} = \sum_{i=0}^{100} \binom{100}{i} x^i \frac{1}{x^{100-i}}$
 $= \sum_{i=0}^{100} \binom{100}{i} x^{2i-100}$. Where k is an even integer

from -100 to 100, the coefficient of x^k with $k = 2i - 100$ is $\binom{100}{\frac{k+100}{2}}$. Otherwise it is 0.

18. WLOG, let $i < n-i$. Then $\frac{\binom{n}{i}}{\binom{n}{i+1}} = \frac{n! (i+1)! (n-i-1)!}{i! (n-i)! n!}$
 $= \frac{i}{n-i} < 1$. By the identity $\binom{n}{i} = \binom{n}{n-i}$,

the other inequalities hold as well.

25. Let $1 \leq k \leq n$.

a) $k \cdot \binom{n}{k}$ counts the number of ways to choose a subset of k people from n , then ~~the~~ "highlighting" one of these people. Another way to count this is to choose the highlighted person from n , then choose the remaining $k-1$ people from $n-1$, getting $n \binom{n-1}{k-1}$.

$$b) k \binom{n}{k} = \frac{k \cdot n!}{k! (n-k)!} = \frac{n \cdot (n-1)!}{(k-1)! (n-k)!} = n \binom{n-1}{k-1}.$$

26a. Let $k \leq r \leq n$.

$\binom{n}{r} \binom{r}{k}$ counts the number of ways to choose a subset of r people and a sub-subset of k people from the r .

If we choose the subsubset first, there are $\binom{n}{k}$ ways, then $\binom{n-k}{r-k}$ ways to finish the r -subset.

28. Let p be prime and $1 \leq k \leq p-1$.

$$\binom{p}{k} = \frac{p!}{k! (p-k)!} \quad \text{Since } p > k \text{ and } p > p-k, \text{ it follows}$$

that $p \mid p!$ but $p \nmid (k! (p-k)!)$, so $p \mid \binom{p}{k}$.

35. Let S be a nonempty set with n elements.

If n is odd, then there is a 1-1 correspondence between even subsets and odd subsets by complementation.

If n is even, fix an element x . Subsets not containing x are in 1-1 correspondence with subsets containing x by adding x . Each pair of sets contains one odd subset and one even subset.

6.5

10. a) $\binom{17}{5}$ 12 croissants + 6 boxes (flavor)

c) $\binom{17}{5}$ assign 2 per kind to start, then we do dots + boxes on the remaining 12 croissants

d) $\binom{28}{4} + \binom{27}{4} + \binom{26}{4}$
 $\uparrow \quad \quad \uparrow \quad \quad \uparrow$
 no broccoli 1 broccoli 2 broccoli

14. $\binom{20}{3}$

20. $\binom{14}{3}$

22. $\binom{27}{5}$ - there are 26 nonflying days, and must insert the 5 flying days amongst them.

6.5

- 32) Since there are 4 I's, 4 S's, 2 P's, and 1 M in MISSISSIPPI, this is the multinomial $\binom{11}{4,4,2,1} = \frac{11!}{4!4!2!1!} = 34,650$.
- 46) (a) If x_i is the number of books on shelf i , then this is the number of nonnegative integer solutions to $x_1 + x_2 + x_3 + x_4 = 12$ which is $\binom{12+4-1}{4-1} = \binom{15}{3} = \frac{15!}{3!12!} = 455$.
- (b) There are 4 ways to place b_1 (one for each shelf), then 5 ways to place b_2 (as the first book in a shelf or the book after b_1), then 6 ways to place b_3 , and so on until there are 15 ways to place b_{12} . In total this gives $15 \cdot 14 \cdot \dots \cdot 5 \cdot 4 = \frac{15!}{3!} = 217,945,728,000$. Alternatively, this is the answer from part (a) times $12!$, the number of ways to permute the 12 books in any given configuration.
- 50) If you have a n distinguishable objects where n_1 are placed in box 1, n_2 are placed in box 2, $\dots$, and n_k are placed in box k , then we give each object a label $1, 2, \dots, n$ and for any assignment of objects we can create the string $a_1 a_2 a_3 \dots a_n$ where $a_i = j$ if the i^{th} object is placed into box j . Any such string is a permutation of the string

$$\underbrace{11 \dots 1}_{n_1 \text{ times}} \underbrace{22 \dots 2}_{n_2 \text{ times}} \dots \underbrace{kk \dots k}_{n_k \text{ times}}$$

which gives us our bijection. E.g. for the specific case $n = 5, n_1 = 2, n_2 = 2, n_3 = 3$, if we label the three boxes A, B, C , placing objects 1, 2, 3, 4, and 5 into these boxes corresponds to permutations of the string $AABBC$.

8.1

- 2a) If $p(n)$ is the number of permutations of a set n , then we can find the number of permutations of a set of $n + 1$ by considering any permutation as a permutation of the first n elements followed by placing the last element anywhere in the $n + 1$ spots before, between, or after the first n elements. This gives $p(n + 1) = (n + 1)p(n)$ as a recursive formula.
- 8) (a) For a bit string of length $n + 1$ that has 3 consecutive 0's, either the first n bits contain 3 consecutive 0's, or the first $n - 3$ bits do not contain 3 consecutive 0's and the string ends with 1000. If $a(n + 1)$ is the number of such strings, the first case contributes $2a(n)$ (by adding a 0 or 1 to the end) and the second case contributes $2^{n-3} - a(n - 3)$ (since there are 2^{n-3} total $n - 3$ length bit strings and $a(n - 3)$ of them contain 3 consecutive 0's). This yields the recursive formula $a(n + 1) = 2a(n) + 2^{n-3} - a(n - 3)$.
- (b) We need the 4 initial conditions $a(0) = a(1) = a(2) = 0, a(3) = 1$.
- (c) Following the recursive formula yields the sequence 0, 0, 0, 1, 3, 8, 20, 47 so $a(7) = 47$.
- 12) (a) If $s(n)$ is the number of ways to climb n stairs 1, 2, or 3 steps at a time, then

$$s(n + 1) = s(n) + s(n - 1) + s(n - 2)$$

by adding to the total sum the contribution from when the penultimate step was 1 down, 2 down, or 3 down from the top.

- (b) We have the initial conditions $s(1) = 1, s(2) = 2, s(3) = 4$.
- (c) Following the recursive formula yields the sequence 1, 2, 4, 7, 13, 24, 44, 81 so $s(8) = 81$.
- 26) (a) If $d(n)$ is the number of ways to cover a $2 \times n$ array with 2×1 dominos, then

$$d(n + 1) = d(n) + d(n - 1)$$

where the first term comes from adding a vertical domino to the last column and the second term comes from adding two horizontal dominos to the last two columns.

- (b) We have the initial conditions $d(1) = 1, d(2) = 2$.

- (c) Following the recursive formula yields the sequence 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, 233, 377, 610, 987, 1597, 2584 so $d(17) = 2, 584$.
- 32) (a) Let $t(n)$ be the number of moves required to solve the puzzle under the restriction. If we know the solution with n disks, we can create the solution for $n + 1$ disks with the following strategy: 1) Move the top n disks from the first peg to the third in $t(n)$ moves; 2) Move the biggest disk to peg 2; 3) Move the n disks back from peg 3 to peg 1 in $t(n)$ moves; 4) Move the biggest disk to peg 3; 4) Move the smallest n disks from the first peg to the third in $t(n)$ moves one last time. This gives the formula $t(n + 1) = 3t(n) + 2$ where $t(1) = 2$.
- (b) Since $t(n + 1) + 1 = 3t(n) + 3 = 3(t(n) + 1)$ for all $n \geq 1$, we have
- $$t(n + 1) + 1 = 3(t(n) + 1) = 3^2(t(n - 1) + 1) = \dots = 3^n(t(1) + 1) = 3^n(2 + 1) = 3^{n+1}$$
- and so $t(n) = 3^n - 1$ for all $n \geq 1$.
- (c) We can calculate valid configurations by placing the n disks consecutively starting from largest to smallest, where there are 3 peg options for every disk, yielding 3^n possible arrangements.
- (d) We know from parts (a) and (b) that there are $3^n - 1$ moves minimum required to solve the puzzle, which means 3^n allowable arrangements occurring in total. These must all be distinct since if there were some repeat, we could solve the puzzle in fewer moves by removing the moves between those 2 occurrences. Therefore all 3^n possible arrangements must be occur in the solution to this variant of the puzzle.
- TC) Let $c(n)$ be the number of 3-colorings of a tree on n -vertices. By induction, for a tree on $n + 1$ vertices, if we delete a leaf then we get a tree on n vertices that can be colored $c(n)$ ways; for any coloring, we can get 2 different colorings of the original tree by coloring the leaf one of the 2 colors not identical to its neighbor's color. Therefore, $c(n + 1) = 2c(n)$ where $c(1) = 3$, and so
- $$c(n) = 2c(n - 1) = 2^2c(n - 2) = \dots = 2^{n-1}c(1) = 3 \cdot 2^{n-1}.$$

6.2

- 6) There are 6 professors and 101 possible scores and therefore 606 possible pairs of professors and scores. We thus need 607 students in order to guarantee the existence of two students with the same professor and score.
- 10) We place the $|S| = n$ items of S into $|T| = k$ buckets based on where they are sent by the function f . By the pigeonhole principle, since $n > k$, there exist two items sent to the same bucket, which corresponds to 2 elements $s_1, s_2 \in S$ with $f(s_1) = f(s_2)$.
- 18) The set $\{1, 3, 5, 7, 9, 11, 13, 15\}$ can be partitioned into the buckets $\{1, 15\}$, $\{3, 13\}$, $\{5, 11\}$, and $\{7, 9\}$; two items in any of these buckets gives us a pair of numbers that sum to 16. Therefore, by the pigeonhole principle, 5 numbers must be selected in order to guarantee that at least one pair fall in the same bucket (i.e. sum to 16).
- 36) We can place the $n = 8, 537, 673$ New Yorkers into $k = 1, 000, 001$ buckets based on the number of hairs on a person's head; since $\lceil \frac{8537673}{1000001} \rceil = 9$, by the generalized pigeonhole principle 9 New Yorkers must be in the same bucket, i.e. have the same number of hairs on their head.
- 47) Let x be an irrational number and n be a positive integer. Our goal is to prove that for some positive integer $j \leq n$, jx is less than $\frac{1}{n}$ away from the nearest integer. Consider the set $S = \{x, 2x, 3x, \dots, nx\}$ of n numbers, all of which are irrational (if $jx = \frac{p}{q}$ for $p, q, j \in \mathbb{Z}$, then $x = \frac{p}{jq}$, a contradiction). We now introduce the function $\text{frac}(x) = x - \lfloor x \rfloor$ which is just the decimal part of a number; for example $\text{frac}(2.786) = .786$ and $\text{frac}(\pi) = .141592\dots$. We now create n buckets where x is in bucket i if $\frac{i-1}{n} < \text{frac}(x) < \frac{i}{n}$. If any elements of S are in the 1^{st} or n^{th} bucket, then we are done since that means it is less than $\frac{1}{n}$ away from the nearest integer. Otherwise, by the pigeonhole principle there exists two elements $kx, lx \in S$ with $0 < k < l \leq n$ so that kx and lx are in the same bucket. Since $\text{frac}((l-k)x) = \text{frac}(lx - kx) = \text{frac}(lx) - \text{frac}(kx) < \frac{1}{n}$, taking $j = l - k$ completes the proof. (Note: in the general case we may not have $\text{frac}(x - y) = \text{frac}(x) - \text{frac}(y)$, but this is true when the right-hand side is in $[0, 1)$.)